

BUA4003 PRINCIPLES OF ECONOMICS

SEMESTER 1 2026

Assessment 1 - Problem Set #1

This assessment is worth 10% of total marks in the subject.

Due date: Friday 20 March 2026 11:59pm

This problem set is related to the following Topics:

- Topic 1 - Introduction: Ten Lessons from Economics
- Topic 2 - The Market Forces of Supply and Demand

You must answer all questions and provide figures where appropriate. The total word count should be 750 words (+/-10%).

Generative AI is allowed in this assessment at **AI Planning** level for planning, idea development and research. The final submission must show how you have developed and refined ideas and must be of your own words. All submissions will be screened via the Turnitin plagiarism software.

Intended Learning Outcomes:

1. Apply economic frameworks for policy and business decision-making.
2. Research and apply economic concepts to predict how changes in economic conditions may impact individuals, businesses and industries within market structures.
3. Construct and present logical and persuasive economic arguments and communicate concepts professionally.

Question 1 (3 marks)

You are the CEO of a mid-sized EV manufacturer looking to enter a new Asia-Pacific market. You are making decision on how to allocate a fixed investment budget of \$500 million across three initiatives below **in priority order**.

- Initiative #1: Scale production of your current mid-range EV model. The projected incremental profit in the next 10 years of this option is \$100 million.
- Initiative #2: Invest in next-gen battery R&D to extend range and lower costs. The projected incremental profit in the next 10 years of this option is \$80 million.
- Initiative #3: Co-invest with partners in urban charging infrastructure. The projected incremental profit in the next 10 years of this option is \$90 million.

In no more than 200 words, discuss the potential reasons why Initiative #2 is ranked higher than Initiative #3; what initiative will you choose and the trade-offs and opportunity cost of your decision.

Question 3 (3 marks)

Suppose that the demand and supply for movie tickets in an economy are given by:

$$P = 50 - 2Q \text{ (demand)}$$

$$P = 20 + Q \text{ (supply)}$$

- Find the equilibrium price and quantity of movie. Show all workings out and explanations. **[1 mark]**
- Using Excel, graph the demand and supply curve and indicate the equilibrium. Make sure you copy the diagram from Excel and include it in the word document submission. **[1 mark]**
- At a price of \$20, find the actual shortage or surplus of movie ticket and indicate this on the graph. **[1 mark]**

Question 4 (4 marks)

Read [this article](#) about the drought in South Australia and Western Victoria. Use the general demand and supply model, illustrate with diagram(s) and explain the impact of this drought on the equilibrium price and quantity of lamb and chicken - a substitute for lamb.